Supplementary Material

A Systematic Review of Influenza Epidemiology and Surveillance in the Eastern Mediterranean and North African Region

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Table S1A: Electronic database search strategy

	Search strategy (Pubmed)	**Results**
S1	"Influenza, Human"[Mesh] OR influenza OR "influenza A" OR "influenza B" OR "seasonal influenza" OR "pandemic influenza" OR flu OR grippe OR H1N1 OR PH1N1 OR H3N2 OR AH1N1 OR AH3N2	101,876
	AND	
S2	"Africa, Northern"[Mesh] OR "Lebanon"[Mesh] OR "Jordan"[Mesh] OR "Djibouti"[Mesh] OR "Yemen"[Mesh] OR "United Arab Emirates"[Mesh] OR "Turkey"[Mesh] OR "Tunisia"[Mesh] OR "Syria"[Mesh] OR "Sudan"[Mesh] OR "Somalia"[Mesh] OR "Saudi Arabia"[Mesh] OR "Qatar"[Mesh] OR "Pakistan"[Mesh] OR "Oman"[Mesh] OR "Morocco"[Mesh] OR "Libya"[Mesh] OR "Kuwait"[Mesh] OR "Iraq"[Mesh] OR "Iran"[Mesh] OR "Egypt"[Mesh] OR "Bahrain"[Mesh] OR "Algeria"[Mesh] OR "Afghanistan"[Mesh] OR "Israel"[Mesh]OR "Middle East"[Mesh]	207,154
	OR	
S3	"Middle East" OR "Middle-East" OR "North Africa" OR "North-Africa" OR "EMRO" OR "Eastern Mediterranean" OR "Arab" OR "Arabs" OR "Arab World" OR "Islam" OR "Afghanistan" OR "Algeria" OR "Bahrain" OR "Djibouti" OR "Egypt" OR "Jordan" OR "Kuwait" OR "Lebanon" OR "Libya" OR "Iran" OR "Iraq" OR "Morocco" OR "Oman" OR "Pakistan" OR "Qatar" OR "Saudi Arabia" OR "Somalia" OR "Sudan" OR "Syria" OR "Tunisia" OR "Turkey" OR "United Arab Emirates" OR "Dubai" OR "Abu Dhabi" OR "Abu-Dhabi" OR "Sharjah" OR "Palestine" "West Bank" OR "Ghaza" OR "Israel" OR "Yemen"	781
S4	Lim 1998−2018	656
	NOT	
S5	Animals[Mesh] NOT (Animals[Mesh] AND "Humans"[Mesh])	
	Total	**589**
	Search strategy (Embase)	
S1	'influenza' OR 'influenza a'/exp OR 'influenza b'/exp OR 'seasonal influenza' OR 'pandemic influenza' OR (influenza* or flu or grippe):ti,ab,kw OR (H1N1 or PH1N1 or H3N2 or AH1N1 or AH3N2):ti,ab,kw	176,851
	AND	
S2	'follow up':ti,ab,kw OR 'follow-up':ti,ab,kw OR 'population research'/exp OR 'population research' OR population:ti,ab,kw OR 'population-based':ti,ab,kw OR 'case-control':ti,ab,kw OR 'case-base':ti,ab,kw OR 'case-based':ti,ab,kw OR 'case-comparison':ti,ab,kw OR 'case-compeer':ti,ab,kw OR 'case-referent':ti,ab,kw OR 'case-referrent':ti,ab,kw OR 'comparative study'/exp OR 'comparative study' OR 'controlled study'/exp OR 'controlled study' OR 'field study'/exp OR 'field study' OR 'experimental study'/exp OR 'experimental study' OR 'observational study'/exp OR 'observational	13,928,312

	study' OR 'pilot study'/exp OR 'pilot study' OR 'prevention study'/exp OR 'prevention study' OR 'trend study'/exp OR 'trend study'	
	OR	
S3	''incidence'/exp OR 'incidence' OR 'prevalence'/exp OR 'prevalence' OR 'disease surveillance'/exp OR 'disease surveillance' OR 'infection rate'/exp OR 'infection rate' OR 'morbidity'/exp OR 'morbidity' OR 'newborn morbidity'/exp OR 'newborn morbidity' OR 'mortality'/exp OR 'mortality' OR 'infant mortality'/exp OR 'infant mortality' OR 'newborn mortality'/exp OR 'newborn mortality' OR 'cause of death'/exp OR 'cause of death' OR 'survival rate'/exp OR 'survival rate' OR burden:ti,ab,kw OR burdens:ti,ab,kw OR death:ti,ab,kw OR deaths:ti,ab,kw OR epidemiolog*:ti,ab,kw OR incidence:ti,ab,kw OR frequenc*:ti,ab,kw OR morbidit*:ti,ab,kw OR mortalit*:ti,ab,kw OR occurrence*:ti,ab,kw OR occurence*:ti,ab,kw OR outbreak*:ti,ab,kw OR prevalen*:ti,ab,kw OR rate:ti,ab,kw OR rates:ti,ab,kw OR surveillance:ti,ab,kw OR 'cohort analysis'/exp OR 'cohort analysis' OR 'cohort':ti,ab,kw OR 'retrospective study'/exp OR 'retrospective study' OR 'longitudinal study'/exp OR 'longitudinal study' OR 'prospective study'/exp OR 'prospective study' OR longitudinal:ti,ab,kw OR prospective:ti,ab,kw OR retrospective:ti,ab,kw	
	AND	
S4	('Middle East' OR 'EMRO' OR 'Eastern Mediterranean' OR 'Arab' OR 'Arab World' OR Islam* OR 'Gulf' OR Afghanistan OR Algeria OR Bahrain OR Djibouti OR Egypt OR Jordan OR Kuwait OR Lebanon OR Libya OR Iran OR Iraq OR Morocco OR Oman OR Pakistan OR Qatar OR 'Saudi Arabia' OR 'KSA' OR 'Kingdom of Saudi Arabia' OR Somalia OR Sudan OR Syria OR Tunisia OR Turkey OR 'United Arab Emirates' OR 'UAE' OR 'Dubai' OR 'Abu Dhabi' OR Sharjah OR 'West Bank' OR Ghaza OR Palestin* OR Israel OR Yemen):ti,ab,kw	1,457,155
	OR	
S5	'Middle East'/exp OR 'North Africa'/exp OR 'Arab'/exp OR 'Afghanistan'/exp OR 'Djibouti'/exp OR 'Pakistan'/exp OR 'Somalia'/exp OR 'Sudan'/exp 'Afghanistan'/exp OR 'Algeria'/exp OR 'Bahrain'/exp OR 'Djibouti'/exp OR 'Egypt'/exp OR 'Jordan'/exp OR 'Kuwait'/exp OR 'Lebanon'/exp OR 'Libya'/exp OR 'Iran'/exp OR 'Iraq'/exp OR 'Morocco'/exp OR 'Oman'/exp OR 'Pakistan'/exp OR 'Qatar'/exp OR 'Saudi Arabia'/exp OR 'Somalia'/exp OR 'Sudan'/exp OR 'Syria'/exp OR 'Tunisia'/exp OR 'Turkey'/exp OR 'United Arab Emirates'/exp OR 'Palestine'/exp OR 'Israel'/exp OR 'Yemen'/exp	
	Total	**3618**
	Search strategy (Scopus)	
S1	' ALL(influenza OR "influenza A" OR "influenza B" OR "seasonal influenza" OR "pandemic influenza" OR "influenza*" OR flu or grippe OR H1N1 OR PH1N1 OR H3N2 OR AH1N1 OR AH3N2)	437,804

	AND	
S2	'TITLE-ABS-KEY("incidence" OR "prevalence" OR "disease surveillance" OR "Infection rate" OR "morbidity" OR "mortality" OR "cause of death" OR "survival rate" OR "burden*" OR "death*" OR "epidemiolog*" OR "incidence" OR "frequenc*" OR "morbidit*" OR "mortalit*" OR "occurrence*" OR "occurence*" OR "outbreak*" OR "prevalen*" OR "rate*" OR "surveillance*")	14,114,978
	OR	
S3	'TITLE-ABS-KEY("cohort analysis" OR "cohort" OR "retrospective study" OR "longitudinal study" OR "prospective study" OR "longitudinal*" OR "prospective" OR "retrospective*" OR (("follow up" W/3 stud*) OR ("follow-up" W/3 stud*)) OR "population research" OR ((population OR "population-based") W/3 (stud* OR analys*)) OR "case-control*" OR "case-base" OR "case-based" OR "case-comparison" OR "case-compeer" OR "case-referent" OR "case-referrent")	
	AND	
S4	'TITLE-ABS-KEY("Middle East" OR "North Africa" OR "Arab" OR "Afghanistan" OR "Djibouti" OR "Pakistan" OR "Somalia" OR "Sudan" OR "EMRO" OR "Eastern Mediterranean" OR "Arab*" OR "Arab World" OR "Islam*" OR "Gulf" OR "Afghanistan" OR "Algeria" OR "Bahrain" OR "Djibouti" OR "Egypt" OR "Jordan" OR "Kuwait" OR "Lebanon" OR "Libya" OR "Iran" OR "Iraq" OR "Morocco" OR "Oman" OR "Pakistan" OR "Qatar" OR "Saudi Arabia" OR "KSA" OR "Kingdom of Saudi Arabia" OR "Somalia" OR "Sudan" OR "Syria" OR "Tunisia" OR "Turkey" OR "Turk*" OR "United Arab Emirates" OR "UAE" or "Dubai" OR "Abu Dhabi" OR "Sharjah" OR "West Bank" OR "Ghaza" OR "Palestin*" OR "Israel" OR "Yemen")	1,039,787
	AND	
S5	'PUBYEAR' > 1997	
	Lim Human - Humans	3435
	Total	**3435**

Table S1B: Updated search strategy

	Search strategy (Pubmed)	**Results**
S1	"Influenza, Human"[Mesh] OR influenza OR "influenza A" OR "influenza B" OR "seasonal influenza" OR "pandemic influenza" OR flu OR grippe OR H1N1 OR PH1N1 OR H3N2 OR AH1N1 OR AH3N2	111,983
	AND	
S2	"Africa, Northern"[Mesh] OR "Lebanon"[Mesh] OR "Jordan"[Mesh] OR "Djibouti"[Mesh] OR "Yemen"[Mesh] OR "United Arab Emirates"[Mesh] OR "Turkey"[Mesh] OR "Tunisia"[Mesh] OR "Syria"[Mesh] OR "Sudan"[Mesh] OR "Somalia"[Mesh] OR "Saudi Arabia"[Mesh] OR "Qatar"[Mesh] OR "Pakistan"[Mesh] OR "Oman"[Mesh] OR "Morocco"[Mesh] OR "Libya"[Mesh] OR "Kuwait"[Mesh] OR "Iraq"[Mesh] OR "Iran"[Mesh] OR "Egypt"[Mesh] OR "Bahrain"[Mesh] OR "Algeria"[Mesh] OR "Afghanistan"[Mesh] OR "Israel"[Mesh]OR "Middle East"[Mesh]	183,929
	OR	
S3	"Middle East" OR "Middle-East" OR "North Africa" OR "North-Africa" OR "EMRO" OR "Eastern Mediterranean" OR "Arab" OR "Arabs" OR "Arab World" OR "Islam" OR "Afghanistan" OR "Algeria" OR "Bahrain" OR "Djibouti" OR "Egypt" OR "Jordan" OR "Kuwait" OR "Lebanon" OR "Libya" OR "Iran" OR "Iraq" OR "Morocco" OR "Oman" OR "Pakistan" OR "Qatar" OR "Saudi Arabia" OR "Somalia" OR "Sudan" OR "Syria" OR "Tunisia" OR "Turkey" OR "United Arab Emirates" OR "Dubai" OR "Abu Dhabi" OR "Abu-Dhabi" OR "Sharjah" OR "Palestine" "West Bank" OR "Ghaza" OR "Israel" OR "Yemen"	234,982
	AND	
S4	Lim 2018−present (Jan 2020)	274
	NOT	
S5	Animals[Mesh] NOT (Animals[Mesh] AND "Humans"[Mesh])	164
	Total	**164**
	Search strategy (Embase)	
S1	influenza' OR 'influenza a'/exp OR 'influenza b'/exp OR 'seasonal influenza' OR 'pandemic influenza' OR (influenza* or flu or grippe):ti,ab,kw OR (H1N1 or PH1N1 or H3N2 or AH1N1 or AH3N2):ti,ab,kw	195,114
	AND	
S2	follow up':ti,ab,kw OR 'follow-up':ti,ab,kw OR 'population research'/exp OR 'population research' OR population:ti,ab,kw OR 'population-based':ti,ab,kw OR 'case-control':ti,ab,kw OR 'case-base':ti,ab,kw OR 'case-based':ti,ab,kw OR 'case-comparison':ti,ab,kw OR 'case-compeer':ti,ab,kw OR 'case-referent':ti,ab,kw OR 'case-referrent':ti,ab,kw OR 'comparative study'/exp OR 'comparative study' OR 'controlled study'/exp OR 'controlled study' OR 'field study'/exp OR 'field study' OR 'experimental study'/exp OR 'experimental study' OR 'observational study'/exp OR 'observational study' OR 'pilot study'/exp OR 'pilot study' OR 'prevention study'/exp OR 'prevention study' OR 'trend study'/exp OR 'trend study'	11,393,813
	OR	
S3	'incidence'/exp OR 'incidence' OR 'prevalence'/exp OR 'prevalence' OR 'disease surveillance'/exp OR 'disease surveillance' OR 'infection rate'/exp OR 'infection rate' OR 'morbidity'/exp	10,734,639

	OR 'morbidity' OR 'newborn morbidity'/exp OR 'newborn morbidity' OR 'mortality'/exp OR 'mortality' OR 'infant mortality'/exp OR 'infant mortality' OR 'newborn mortality'/exp OR 'newborn mortality' OR 'cause of death'/exp OR 'cause of death' OR 'survival rate'/exp OR 'survival rate' OR burden:ti,ab,kw OR burdens:ti,ab,kw OR death:ti,ab,kw OR deaths:ti,ab,kw OR epidemiolog*:ti,ab,kw OR incidence:ti,ab,kw OR frequenc*:ti,ab,kw OR morbidit*:ti,ab,kw OR mortalit*:ti,ab,kw OR occurrence*:ti,ab,kw OR occurence*:ti,ab,kw OR outbreak*:ti,ab,kw OR prevalen*:ti,ab,kw OR rate:ti,ab,kw OR rates:ti,ab,kw OR surveillance:ti,ab,kw OR 'cohort analysis'/exp OR 'cohort analysis' OR 'cohort':ti,ab,kw OR 'retrospective study'/exp OR 'retrospective study' OR 'longitudinal study'/exp OR 'longitudinal study' OR 'prospective study'/exp OR 'prospective study' OR longitudinal:ti,ab,kw OR prospective:ti,ab,kw OR retrospective:ti,ab,kw	
	AND	
S4	('Middle East' OR 'EMRO' OR 'Eastern Mediterranean' OR 'Arab' OR 'Arab World' OR Islam* OR 'Gulf' OR Afghanistan OR Algeria OR Bahrain OR Djibouti OR Egypt OR Jordan OR Kuwait OR Lebanon OR Libya OR Iran OR Iraq OR Morocco OR Oman OR Pakistan OR Qatar OR 'Saudi Arabia' OR 'KSA' OR 'Kingdom of Saudi Arabia' OR Somalia OR Sudan OR Syria OR Tunisia OR Turkey OR 'United Arab Emirates' OR 'UAE' OR 'Dubai' OR 'Abu Dhabi' OR Sharjah OR 'West Bank' OR Ghaza OR Palestin* OR Israel OR Yemen):ti,ab,kw	1,797,008
	OR	
S5	Middle East'/exp OR 'North Africa'/exp OR 'Arab'/exp OR 'Afghanistan'/exp OR 'Djibouti'/exp OR 'Pakistan'/exp OR 'Somalia'/exp OR 'Sudan'/exp OR 'Afghanistan'/exp OR 'Algeria'/exp OR 'Bahrain'/exp OR 'Djibouti'/exp OR 'Egypt'/exp OR 'Jordan'/exp OR 'Kuwait'/exp OR 'Lebanon'/exp OR 'Libya'/exp OR 'Iran'/exp OR 'Iraq'/exp OR 'Morocco'/exp OR 'Oman'/exp OR 'Pakistan'/exp OR 'Qatar'/exp OR 'Saudi Arabia'/exp OR 'Somalia'/exp OR 'Sudan'/exp OR 'Syria'/exp OR 'Tunisia'/exp OR 'Turkey'/exp OR 'United Arab Emirates'/exp OR 'Palestine'/exp OR 'Israel'/exp OR 'Yemen'/exp	265,250
	AND	
S6	Lim 2018-2020	1,331
S7	Lim Human	961
	Total	**961**
	Search strategy (Scopus)	
S1	ALL(influenza OR "influenza A" OR "influenza B" OR "seasonal influenza" OR "pandemic influenza" OR "influenza*" OR flu or grippe OR H1N1 OR PH1N1 OR H3N2 OR AH1N1 OR AH3N2)	499,495
	AND	
S2	TITLE-ABS-KEY("incidence" OR "prevalence" OR "disease surveillance" OR "Infection rate" OR "morbidity" OR "mortality" OR "cause of death" OR "survival rate" OR "burden*" OR "death*" OR "epidemiolog*" OR "incidence" OR "frequenc*" OR "morbidit*" OR "mortalit*" OR "occurrence*" OR "occurence*" OR "outbreak*" OR "prevalen*" OR "rate*" OR "surveillance*")	8,919,298
	OR	
S3	TITLE-ABS-KEY("cohort analysis" OR "cohort" OR "retrospective study" OR "longitudinal study" OR "prospective study" OR "longitudinal*" OR "prospective" OR "retrospective*" OR (("follow up" W/3 stud*) OR	277,597

	("follow-up" W/3 stud*)) OR "population research" OR ((population OR "population-based") W/3 (stud* OR analys*)) OR "case-control*" OR "case-base" OR "case-based" OR "case-comparison" OR "case-compeer" OR "case-referent" OR "case-referrent")	
	AND	
S4	TITLE-ABS-KEY("Middle East" OR "North Africa" OR "Arab" OR "Afghanistan" OR "Djibouti" OR "Pakistan" OR "Somalia" OR "Sudan" OR "EMRO" OR "Eastern Mediterranean" OR "Arab*" OR "Arab World" OR "Islam*" OR "Gulf" OR "Afghanistan" OR "Algeria" OR "Bahrain" OR "Djibouti" OR "Egypt" OR "Jordan" OR "Kuwait" OR "Lebanon" OR "Libya" OR "Iran" OR "Iraq" OR "Morocco" OR "Oman" OR "Pakistan" OR "Qatar" OR "Saudi Arabia" OR "KSA" OR "Kingdom of Saudi Arabia" OR "Somalia" OR "Sudan" OR "Syria" OR "Tunisia" OR "Turkey" OR "Turk*" OR "United Arab Emirates" OR "UAE" or "Dubai" OR "Abu Dhabi" OR "Sharjah" OR "West Bank" OR "Ghaza" OR "Palestin*" OR "Israel" OR "Yemen")	1,220,180
	AND	
S5	PUBYEAR > 1997	1023
S6	Lim Human - Humans	641
	Total	**641**

Table S2: **Data extraction form**

Study characteristics	Author
	Year of publication
	Country
	Study design
	Study period and duration
	Study setting
	Pandemic 2009
Study population	Type
	Mean (median) age (SD, var)
	% of Females
	Comorbidities
	% of vaccinated
Clinical/virology	Case definition
	Biofluid collection method
	Laboratory method
	Strains
	Other viruses
Epidemiology	N of enrolled patients
	N (%) flu positive
	N (%) A strain
	N (%) B strain
	Incidence of influenza
	Prevalence
	Hospitalisation due to influenza
	Mortality due to influenza
	DALY (mean/median)
	QALY (mean/median)

DALY, disability-adjusted life year; N, number; QALY, quality-adjusted life year; SD, standard deviation; var, variance